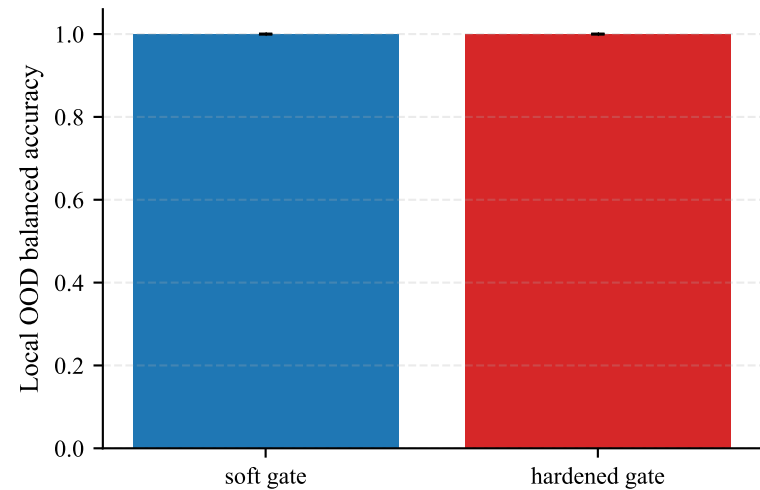
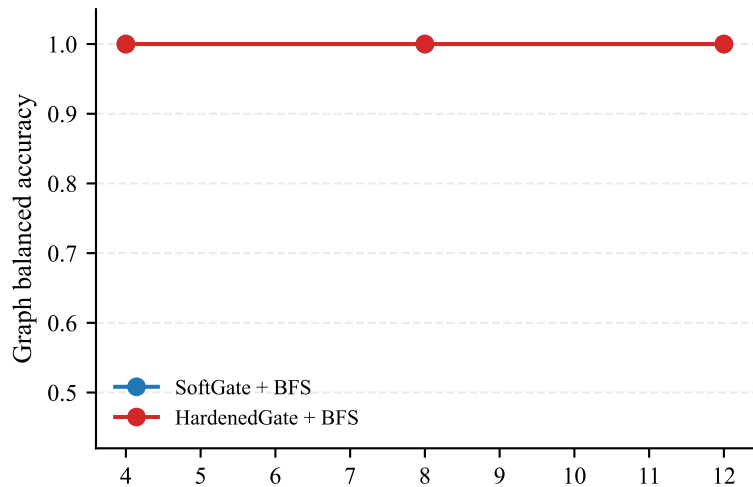




(a) fixed-wiring hardening



(b) path length